

WIEN UNTERLAA, Austria

WMO: 110400

Lat: 48.125N

Lon: 16.419E

Elev: 200

StdP: 98.95

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.5	-7.2	-13.9	1.1	-8.5	-11.2	1.5	-5.8	14.0	7.1	12.7	5.6	3.1	330	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	32.1	21.5	30.2	20.8	28.5	20.0	22.3	29.9	21.4	28.6	20.6	27.1	3.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.8	25.6	19.0	14.1	25.0	18.2	13.4	24.2	66.6	30.0	63.2	28.8	60.2	27.3	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.3	8.2	-11.5	34.8	3.0	2.0	-13.6	36.2	-15.4	37.4	-17.0	38.5	-19.2	40.0		
(5)			-12.2	23.6	2.9	1.0	-14.3	24.3	-15.9	24.9	-17.6	25.4	-19.6	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	0.6	2.2	6.4	12.0	15.7	19.9	21.8	21.3	16.4	11.1	6.2	1.6		
	DBStd	8.43	4.58	4.70	4.50	3.84	3.62	3.49	3.26	3.25	3.40	3.80	3.81	4.00		
	HDD10.0	1083	294	220	126	24	0	0	0	0	1	34	123	259		
	HDD18.3	2887	551	451	369	193	97	24	8	9	78	226	363	518		
	CDD10.0	1561	1	2	14	83	178	296	367	352	192	67	11	1		
	CDD18.3	323	0	0	0	2	14	70	116	102	18	1	0	0		
	CDH23.3	2604	0	0	0	21	107	538	977	835	123	4	0	0		
	CDH26.7	824	0	0	0	2	15	146	342	297	23	0	0	0		
(14) Wind	WSAvg	3.8	4.3	4.1	4.2	4.0	3.9	3.5	3.6	3.2	3.4	3.4	3.8	3.9		
(15) Precipitation	PrecAvg	649	35	32	48	43	71	78	78	70	66	45	44	38		
	PrecMax	865	87	74	119	99	168	179	226	169	201	97	86	69		
	PrecMin	433	3	0	14	1	18	23	8	15	20	4	0	11		
	PrecStd	112	20	22	28	27	36	38	43	40	41	24	19	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.7	15.9	20.3	26.0	28.6	32.4	34.3	34.8	29.3	23.7	18.1	13.4		
		MCWB	9.1	9.1	12.3	16.5	19.8	21.9	22.2	21.6	19.5	16.3	13.1	9.6		
	2%	DB	11.2	13.5	17.8	22.9	26.2	30.2	32.2	32.2	26.6	20.6	15.6	10.8		
		MCWB	7.7	8.7	11.0	15.0	18.5	21.2	21.6	21.2	18.5	15.2	11.9	7.6		
	5%	DB	9.0	11.0	15.4	20.7	24.2	28.3	30.2	29.9	24.3	18.6	13.6	8.6		
		MCWB	6.3	7.3	9.8	13.6	17.4	20.2	20.8	20.6	17.5	14.2	10.7	6.1		
	10%	DB	6.9	8.8	13.3	18.8	22.3	26.5	28.4	27.8	22.3	16.9	11.6	7.0		
		MCWB	4.6	5.9	8.6	12.6	16.0	19.1	20.0	19.7	16.5	13.3	9.3	4.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	10.2	12.9	17.1	20.5	23.5	23.7	23.2	20.2	17.1	13.7	9.7		
		MCDB	13.0	14.7	19.1	24.7	27.2	29.9	31.4	32.2	28.0	21.5	16.8	12.8		
	2%	WB	7.8	8.6	11.4	15.4	19.0	22.1	22.3	22.0	19.0	15.8	12.5	8.0		
		MCDB	10.7	12.7	16.7	22.2	25.3	28.5	30.1	29.9	25.1	19.7	15.2	10.6		
	5%	WB	6.3	7.4	10.3	14.1	17.8	20.8	21.4	21.1	18.1	14.7	10.9	6.2		
		MCDB	8.7	10.8	14.8	19.9	23.1	26.7	28.8	28.3	23.4	17.7	13.3	8.2		
	10%	WB	4.7	6.0	9.0	13.0	16.7	19.8	20.6	20.2	17.1	13.7	9.5	5.1		
		MCDB	6.7	8.6	12.9	18.1	21.3	25.4	27.2	26.5	21.4	16.3	11.4	6.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.7	7.9	9.3	8.8	9.3	9.6	9.6	8.3	7.1	4.9	4.5		
		MCDBR	7.4	9.0	12.2	12.8	12.1	12.2	12.6	13.3	12.0	10.3	7.4	7.5		
	5% WB	MCWBR	5.2	5.7	7.0	6.5	5.8	5.0	4.6	4.8	5.1	5.5	4.6	5.3		
		MCWBR	7.1	8.5	11.5	12.1	11.4	11.6	11.7	12.3	11.2	9.5	6.9	7.1		
(40) Clear-Sky Solar Irradiance	taub		0.319	0.343	0.382	0.427	0.425	0.430	0.437	0.436	0.408	0.377	0.349	0.318		
		taud	2.379	2.338	2.259	2.164	2.215	2.240	2.223	2.240	2.286	2.344	2.377	2.405		
	Ebn at Noon		745	806	826	822	839	836	823	805	790	753	696	698		
		Edh at Noon	70	90	114	138	136	134	134	127	110	89	70	61		
(44) All-Sky Solar Radiation	RadAvg		0.99	1.81	2.98	4.48	5.28	5.75	5.56	4.96	3.54	2.16	1.12	0.78		
	RadStd		0.12	0.24	0.39	0.45	0.56	0.43	0.51	0.48	0.43	0.24	0.11	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (85 neighbors)	+0.56	N/A	N/A	+0.83	+0.45	+0.38	-120	-160	+85	+43

Nomenclature: See separate page